# Human nutrition

Mark scheme and worked solutions for 30 questions: 15 multiple choice, 15 structured.

## 1 · Part 1

Core · Diet · [1]

**Answer: B**

### Worked solution

Match each nutrient with both its source and its importance: iron comes from red meat and green vegetables and is needed to make haemoglobin; vitamin C comes from citrus fruits and is needed for healthy skin and gums; fibre comes from wholemeal cereals, fruit and vegetables and prevents constipation. Only the vitamin D row is correct in both columns.

## 2 · Part 1

Core · Diet · [1]

**Answer: D**

### Explanation

Soft, weak bones that bend are signs of rickets, which is caused by a lack of vitamin D (or calcium) in the diet. Lack of vitamin C causes scurvy, lack of iron reduces the making of haemoglobin rather than softening bones, and too much fat does not cause bones to soften.

## 3 · Part 1

Core · Digestive system · [1]

**Answer: A**

### Explanation

Bile is made in the liver and stored in the gall bladder until it is released into the duodenum. The liver makes bile but does not store it. The pancreas and salivary glands secrete digestive enzymes, not bile.

## 4 · Part 1

Core · Physical digestion · [1]

**Answer: C**

### Explanation

Physical digestion breaks food into smaller pieces without any chemical change to the food molecules, for example by chewing. Option A and option B both describe chemical digestion (enzymes break bonds to make new, smaller molecules). Option D describes ingestion.

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## 5 · Part 1 Core Physical digestion [1]

**Answer: A**

**Explanation**

Incisors are the chisel-shaped front teeth used for cutting and biting off pieces of food. Canines grip and tear food, not grind it. Molars and premolars grind and crush food; they are not used for biting off pieces.

## 6 · Part 1 Core Physical digestion [1]

**Answer: B**

**Explanation**

The crown is covered by hard enamel; beneath this is dentine; the pulp cavity, with nerves and blood vessels, is in the centre. The other lists put enamel or pulp in the wrong position.

## 7 · Part 1 Core Physical digestion [1]

**Answer: D**

**Explanation**

Physical digestion means breaking food into smaller pieces without chemical change. The stomach does this by contracting the muscles in its wall to churn the food. Options A and B are true functions of substances in gastric juice, but secreting protease is chemical digestion and acid killing microorganisms is not digestion at all. Option C is wrong because most water is absorbed in the small intestine, not the stomach.

## 8 · Part 1 Extended Physical digestion [1]

**Answer: C**

**Worked solution**

Think of bile as a detergent rather than an enzyme. The fat molecules are unchanged, but they are spread out into tiny droplets, so lipase molecules can reach far more fat at once and chemical digestion is faster.

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## 9 · Part 1 Core Chemical digestion [1]

**Answer: B**

**Explanation**

Lipase breaks down fats and oils into fatty acids and glycerol. Row A has amylase acting on protein, row C has protease acting on starch, and row D gives amino acids as the product of starch digestion; amylase breaks starch down to simple reducing sugars and proteases break protein down to amino acids.

## 10 · Part 1 Core Chemical digestion [1]

**Answer: A**

**Explanation**

Lipase is secreted by the pancreas and acts in the small intestine. The liver secretes bile, not lipase (row C). Rows B and D place the action of lipase in the stomach, but lipase acts in the small intestine, and row B also places its secretion in the wrong organ.

## 11 · Part 1 Core Chemical digestion [1]

**Answer: D**

**Explanation**

One function of hydrochloric acid in gastric juice is to kill harmful microorganisms in food, so without it more would survive. Hydrochloric acid does not emulsify fats (option A); bile does that, in the small intestine. Protein digestion in the stomach would be slower, not faster, because the stomach protease works best in acidic conditions (option B). Hydrochloric acid has no role in water absorption (option C).

## 12 · Part 1 Extended Chemical digestion [1]

**Answer: C**

**Explanation**

Amylase breaks down starch to maltose, and maltase then breaks down maltose to glucose. Row A has the two enzymes the wrong way round. Proteases digest protein (row B) and lipase digests fats and oils (row D), so neither acts on starch or maltose.

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## 13 · Part 1 Extended Chemical digestion [1]

**Answer: A**

### Worked solution

Match each protease to where it normally acts. Stomach (pepsin) is acidic, so pepsin needs a low pH. Small intestine (trypsin) is alkaline, so trypsin needs a high pH. The fastest tubes are the ones where each enzyme is at its own suitable pH.

## 14 · Part 1 Core Absorption [1]

**Answer: B**

### Explanation

The small intestine is the region where nutrients such as glucose and amino acids are absorbed into the blood. The stomach and oesophagus do not absorb nutrients, and the colon absorbs some water but not nutrients.

## 15 · Part 1 Extended Absorption [1]

**Answer: D**

### Explanation

Microvilli are tiny folds of the cell membrane of the epithelial cells, and they greatly increase the internal surface area for absorption. Mitochondria release energy but do not increase surface area. The nucleus and the partially permeable membrane are present in every cell and do not increase the surface area of the intestine.

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## 16 · Part 1 Core Diet [5]

**Mark scheme**

**(a)** **[2]**

- (a diet containing) all the nutrients / named nutrient groups needed (1)
- in the correct proportions / amounts (1)

**ALLOW** provides the right amount of energy for the person's needs, for the second mark

**(b)(i)** **[1]**

- wholemeal / wholegrain bread or cereals / fruit / vegetables / beans or lentils (1)

**REJECT** white bread

**(b)(ii)** **[1]**

- prevents constipation / helps food (and faeces) to move along the alimentary canal (1)

**(c)** **[1]**

- any one from: acts as a solvent / dissolves substances / transports substances (in blood) / needed for chemical reactions in cells / replaces water lost (e.g. in sweat and urine) / helps control body temperature (1)

**Worked solution**

(a) A balanced diet does not mean "lots of every food". It means getting every nutrient you need - carbohydrates, fats and oils, proteins, vitamins, mineral ions, fibre and water - in the right amounts for your age and activity.

(b) White bread has had most of the fibre removed, so Leon needs wholemeal bread, fruit or vegetables. Fibre is not digested; it gives bulk to the food so it moves through the gut easily, preventing constipation.

(c) Water is the solvent in which substances are dissolved and transported, for example in blood plasma.

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## 17 · Part 1 Core Diet [6]

**Mark scheme**

**(a)** **[2]**

- scurvy (1)
- lack of vitamin C (in the diet) (1)

**(b)** **[3]**

- vitamin C from potato = 120 × 20 ÷ 100 = 24 mg (1)
- vitamin C still needed = 40 − 24 = 16 mg (1)
- mass of orange = 16 ÷ 50 × 100 = 32 g (1)

**ECF** final mark for (40 − their potato value) ÷ 50 × 100 correctly evaluated

- correct answer with no working scores full marks

**(c)** **[1]**

- oily fish / eggs or egg yolk / milk or dairy products / liver / fortified foods such as margarine or cereals (1)

**IGNORE** sunlight (not a dietary source)

**Worked solution**

(a) Tinned meat, rice, pasta and bread contain very little vitamin C. Fresh fruit and vegetables are the main sources, so without them the scientists would develop scurvy, which is caused by a lack of vitamin C.

(b) Potato has 20 mg in every 100 g, so 120 g gives 1.2 × 20 = 24 mg. The scientist still needs 40 − 24 = 16 mg. Orange has 50 mg per 100 g, which is 0.5 mg per gram, so 16 ÷ 0.5 = 32 g of orange. Check: 32 g of orange gives 16 mg, and 16 + 24 = 40 mg.

(c) Vitamin D is found in oily fish, egg yolk and dairy products.

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## 18 · Part 1 Core Digestive system [6]

**Mark scheme**

**(a)(i)** **[1]**

- assimilation (1)

**(a)(ii)** **[1]**

- egestion (1)

**REJECT** excretion

**(a)(iii)** **[1]**

- absorption (1)

**(b)** **[1]**

- carries / moves food (and drink) from the mouth (pharynx) to the stomach (1)

**(c)** **[2]**

- fibre is not broken down into small (soluble) molecules (1)
- (fibre molecules are too large / insoluble so) cannot pass through the wall of the intestine into the blood (1)

**Worked solution**

(a) Assimilation is the uptake and use of nutrients by cells, such as muscle cells building proteins from amino acids. Egestion is the removal of undigested food as faeces; do not confuse it with excretion, which removes waste made by cells. Absorption is the movement of nutrients from the intestines into the blood.

(b) The oesophagus is the tube that carries swallowed food from the mouth to the stomach.

(c) Only small, soluble molecules can be absorbed. Fibre is never chemically digested, so it stays as large, insoluble molecules that cannot cross the intestine wall. It passes through and is egested in the faeces.

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## 19 · Part 1 Core Physical digestion [6]

**Mark scheme**

**(a)(i)** **[1]**

- pulp (1)

**ALLOW** pulp cavity

**(a)(ii)** **[1]**

- cement (1)

**ALLOW** cementum

**(b)(i)** **[1]**

- gripping / holding / tearing food (1)

**(b)(ii)** **[1]**

- grinding / crushing / chewing food (1)

**(c)** **[2]**

- chewing breaks food into smaller pieces, increasing the surface area (1)
- (more surface) for enzymes / amylase to act on, so chemical digestion is faster (1)

**Worked solution**

(a) The pulp in the centre of the tooth contains nerves and blood vessels. Cement covers the root and helps hold the tooth in the jaw bone.

(b) Canines are pointed and used to grip and tear food. Premolars (like molars) have broad surfaces with ridges for grinding and crushing food.

(c) Chewing is physical digestion. It does not change the food molecules, but smaller pieces have a much larger total surface area, so enzymes such as amylase in saliva can reach more of the food and digest it faster.

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## 20 · Part 1 Extended Physical digestion [6]

**Mark scheme**

**(a)** **[1]**

- fatty acids and glycerol (both needed) (1)

**(b)** **[1]**

- fatty acids (produced by digestion) are acidic / lower the pH (1)

**(c)** **[2]**

- (15 − 6) ÷ 15 × 100 or 9 ÷ 15 × 100 (1)
- 60 (%) (1)
- correct answer with no working scores full marks

**(d)** **[2]**

- bile salts emulsify the fat / break large fat droplets into smaller droplets (1)
- increasing the surface area for lipase to act on, so fat is digested faster (1)

**REJECT** bile salts digest fat / bile salts are enzymes

**Worked solution**

(a) Lipase digests fat into fatty acids and glycerol.

(b) Fatty acids are acids, so as they are produced the pH falls.

(c) The time decreased by 15 − 6 = 9 minutes. As a percentage of the original time, 9 ÷ 15 × 100 = 60%. Check: 60% of 15 is 9, and 15 − 9 = 6 minutes.

(d) Bile salts are not enzymes, so they do not digest fat themselves. They emulsify the fat into tiny droplets. This gives lipase a much larger surface area to work on, so fatty acids are produced faster and the pH falls in less time.

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## 21 · Part 1 Core Chemical digestion [4]

**Mark scheme**

**(a)** **[2]**

- breakdown of large (insoluble) molecules (1)
- into small (soluble) molecules (1)

> **ALLOW** breakdown of food molecules by enzymes into smaller molecules, for one mark only if size/solubility not mentioned

**(b)** **[1]**

- so that they can be absorbed (into the blood) (1)

**(c)** **[1]**

- amino acids (1)

**Worked solution**

(a) Chemical digestion is the breakdown of large insoluble molecules, such as starch and protein, into small soluble molecules, using enzymes.

(b) Only small soluble molecules can pass through the wall of the small intestine into the blood, so digestion must happen before absorption.

(c) Proteases break protein down into amino acids.

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## 22 · Part 1 Core Chemical digestion [7]

**Mark scheme**

**(a)** **[3]**

- gap 1: salivary glands (1)
- gap 2: mouth (1)
- gap 4: stomach (1)

**ALLOW** buccal cavity for mouth

**(b)** **[2]**

- gap 3: simple reducing sugars (1)

**ALLOW** maltose

**REJECT** glucose

- gap 5: protein (1)

**(c)** **[2]**

- the stomach is acidic / has a low pH / contains hydrochloric acid (1)
- (amylase is) not at its optimum pH / is denatured, so it no longer breaks down starch (1)

**Worked solution**

(a) Amylase is secreted by the salivary glands and the pancreas, and acts in the mouth and the small intestine. Protease is secreted by the stomach and the pancreas.

(b) Amylase breaks starch down into simple reducing sugars (maltose). Proteases break protein down into amino acids.

(c) Saliva is close to neutral, which suits amylase. The hydrochloric acid in the stomach makes the pH very low. This is far from the optimum pH for amylase, so the enzyme is denatured and starch digestion stops until the food reaches the small intestine.

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## 23 · Part 1 Core Chemical digestion [5]

**Mark scheme**

**(a)** **[1]**

- hydrochloric acid (1)

**(b)** **[2]**

- pH rises (from 2.0) to 5.0 in the first 15 minutes (after the meal) (1)
- then falls gradually / steadily back to 2.0 by 120 minutes (1)

**ALLOW** correctly quoted data for each trend

**(c)** **[2]**

- the pH is high(er) / less acidic (at pH 5.0) (1)
- (so) protease is not at its optimum pH / works best in acidic conditions, so its activity is low (1)

**ALLOW** pepsin for protease

**Worked solution**

(a) Gastric juice contains hydrochloric acid.

(b) Before the meal the stomach contents are very acidic (pH 2.0). Eating food raises the pH to 5.0 by 15 minutes. As more gastric juice is secreted, the pH falls steadily - 4.0 at 30 minutes, 3.0 at 60 minutes, 2.5 at 90 minutes - and returns to 2.0 at 120 minutes.

(c) One function of the hydrochloric acid is to provide an acidic pH for optimum enzyme activity. The protease in the stomach works best in strongly acidic conditions. At pH 5.0 it is well away from its optimum, so it digests protein slowly until the acid builds up again.

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## 24 · Part 1 Extended Chemical digestion [7]

**Mark scheme**

**(a)** **[1]**

- maltose (1)

**(b)** **[2]**

- maltase (1)
- on the membranes of the epithelium lining the small intestine (1)

**ALLOW** on the membranes of the epithelial cells / cells lining the small intestine

**REJECT** in the lumen of the small intestine / in the stomach

**(c)** **[2]**

- less glucose absorbed (1)
- because less maltose is broken down to glucose / maltose remains undigested (1)

**(d)** **[2]**

- starch molecules are large / insoluble (1)
- so cannot pass through the wall of the small intestine (into the blood) / only small soluble molecules such as glucose can be absorbed (1)

**Worked solution**

(a) Amylase breaks starch down into maltose.

(b) Maltase breaks maltose down into glucose. It acts on the membranes of the epithelium lining the small intestine, not in the stomach and not free in the intestine contents.

(c) With less maltase, less maltose is converted to glucose. Maltose is not the form in which the sugar is absorbed, so less glucose enters the blood after a starchy meal.

(d) Starch is a large, insoluble molecule. It cannot pass through the epithelium of the small intestine into the blood. Digestion turns it into small, soluble glucose molecules that can be absorbed.

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## 25 · Part 1 Extended Chemical digestion [7]

**Mark scheme**

**(a)** **[2]**

- (enzyme) P (1)
- (works only / best at) low pH / acidic pH / optimum at pH 2, (which matches the acidic conditions in the stomach) (1)
- no mark for P without a reason

**(b)** **[3]**

- bile is alkaline (1)
- neutralises the acid(ic mixture) from the stomach (1)
- provides a suitable / optimum pH for enzyme Q (about pH 8) (1)

**ALLOW** raises the pH for the third mark if linked to enzyme Q working

**(c)** **[2]**

- the contents of the duodenum stay acidic / the acid is not neutralised (1)
- (so) enzyme Q / trypsin works slowly or not at all, e.g. rate of 0 to 2 arbitrary units at pH 2 to 4 compared with 10 at pH 8, so less protein is digested (1)

**Worked solution**

(a) Enzyme P has its highest rate at pH 2 and does not work at all from pH 6 upwards. This is the acidic pH of the stomach, so P is pepsin. (Enzyme Q, with its optimum at pH 8, behaves like trypsin.)

(b) The food leaving the stomach is acidic. Bile is an alkaline mixture, so when it enters the duodenum it neutralises this acid. This raises the pH to a suitable value for enzyme Q, which works best at about pH 8.

(c) Without bile, the acid from the stomach is not neutralised, so the duodenum stays at a low pH. The table shows that enzyme Q has a rate of only 0 to 2 arbitrary units between pH 1 and pH 4, compared with 10 at pH 8, so much less protein would be digested.

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## 26 · Part 1 Core Absorption [5]

**Mark scheme**

**(a)** **[1]**

- small intestine (1)

**ALLOW** ileum

**(b)** **[2]**

- 7.2 ÷ 9.0 × 100 (1)
- 80 (%) (1)
- correct answer with no working scores full marks

**(c)** **[1]**

- 0.2 $dm^3$ (1)

**ALLOW** 200 $cm^3$

**(d)** **[1]**

- faeces are more watery / runny / diarrhoea (1)

**Worked solution**

(a) Nutrients are absorbed in the small intestine, and most water is absorbed there too, so region R is the small intestine.

(b) 7.2 $dm^3$ out of 9.0 $dm^3$ is absorbed in region R: 7.2 ÷ 9.0 × 100 = 80%. Check: 80% of 9.0 is 0.8 × 9.0 = 7.2.

(c) Total absorbed is 7.2 + 1.6 = 8.8 $dm^3$, so 9.0 − 8.8 = 0.2 $dm^3$ leaves in the faeces.

(d) The colon absorbs some of the remaining water. If it absorbs less, more water is left in the faeces, so they become watery.

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## 27 · Part 1 Extended Absorption [7]

**Mark scheme**

**(a)** **[3]**

- villi and microvilli (greatly) increase the (internal) surface area (1)
- so more nutrients / digested food can be absorbed (1)
- faster / at a greater rate (so absorption is completed before food leaves the small intestine) (1)

**ALLOW** reference to more diffusion or active transport into the blood for the third mark

**(b)** **[2]**

- $3.0 \times 8 \times 20$ (1)
- 480 $cm^2$ (1)

**ALLOW** one mark for 24 $cm^2$ or 60 $cm^2$ (only one of the two factors applied)

**(c)** **[2]**

- mitochondria release energy (by aerobic respiration) (1)
- for active transport (of nutrients such as glucose into the cells / against a concentration gradient) (1)

**Worked solution**

(a) Villi are finger-like projections of the lining, and microvilli are tiny folds on the epithelial cells of each villus. Together they give a very large internal surface area, so far more nutrients can be absorbed into the blood, and absorption is fast enough to be finished before the food leaves the small intestine.

(b) Villi multiply the area by 8: $3.0 \times 8 = 24$ $cm^2$. Microvilli multiply this by 20: $24 \times 20 = 480$ $cm^2$. Check backwards: $480 \div 20 = 24$ and $24 \div 8 = 3.0$ $cm^2$.

(c) Some nutrients, such as glucose, are absorbed by active transport, which moves them against a concentration gradient and needs energy. Mitochondria are where aerobic respiration releases this energy, so many mitochondria provide plenty of energy for active transport.

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## 28 · Part 1 Extended Absorption [5]

**Mark scheme**

**(a)** **[4]**

- any four from:
- finger-like projection (of the lining of the small intestine) (1)
- covered by an epithelium (wall) only one cell thick (1)
- epithelial cells have microvilli (1)
- (dense) network of (blood) capillaries (1)
- (a) lacteal in the centre (1)

**ALLOW** thin wall for one cell thick

**(b)** **[1]**

- short distance for nutrients to move / diffuse (into the blood) so absorption is faster (1)

**Worked solution**

(a) A villus is a finger-like projection from the inner wall of the small intestine. It is covered by a single layer of epithelial cells, and each of these cells has microvilli on its surface. Inside the villus is a dense network of blood capillaries, and in the centre is a lacteal.

(b) Because the epithelium is only one cell thick, nutrients only have a very short distance to travel to reach the capillaries and lacteal, so they are absorbed quickly.

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## 29 · Part 1 Extended Absorption [5]

**Mark scheme**

**(a)(i)** **[1]**

- (blood) capillaries (1)

**(a)(ii)** **[1]**

- lacteal (1)

**(b)** **[2]**

- blood carries absorbed glucose away (from the villus) (1)
- keeping the glucose concentration in the blood low / maintaining a concentration gradient, so (diffusion of) glucose into the blood continues (1)

**(c)** **[1]**

- less fat / fewer fatty acids and glycerol absorbed / more fat in the faeces (1)

**ALLOW** loses mass / energy from fats not obtained

**Worked solution**

(a) Glucose and amino acids are absorbed into the blood capillaries of the villus. Fatty acids and glycerol are absorbed into the lacteal.

(b) Once glucose has entered the blood, the moving blood carries it away from the villus. This keeps the concentration of glucose in the blood lower than in the small intestine, so the concentration gradient is kept and glucose continues to move into the blood.

(c) If the lacteals are blocked, fatty acids and glycerol cannot be absorbed, so the person gets less energy from fats in the diet and more fat is left in the faeces.

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## 30 · Part 1 Extended Absorption [7]

**Mark scheme**

**(a)** **[2]**

- 5.70 ÷ (0.30 + 5.70) × 100 or 5.70 ÷ 6.00 × 100 (1)
- 95 (%) (1)
- correct answer with no working scores full marks

**(b)** **[3]**

- the ileum is (much) longer than the colon / 5.70 m compared with 1.50 m, giving a larger surface area / more time for absorption (1)
- the ileum has villi but the colon does not (1)
- villi (and microvilli) (greatly) increase the surface area for absorption (1)

**ALLOW** villi have capillaries / lacteals to carry nutrients away, for the third mark

**(c)** **[2]**

- any two from:
- fewer nutrients / less digested food absorbed (1)
- loss of body mass / malnutrition / deficiency of a named nutrient (1)
- more undigested or unabsorbed food in the faeces / more watery faeces (1)

**Worked solution**

(a) Total length of small intestine = 0.30 + 5.70 = 6.00 m. Percentage that is ileum = 5.70 ÷ 6.00 × 100 = 95%. Check: 5% of 6.00 m is 0.30 m, which is the length of the duodenum.

(b) The ileum is 5.70 m long, nearly four times the length of the colon, so food spends longer there and there is more lining to absorb through. It also has villi, which the colon does not, and the villi and their microvilli give a huge internal surface area. Each villus also has capillaries and a lacteal to carry absorbed nutrients away.

(c) With less ileum, there is less surface area for absorption, so fewer nutrients are absorbed. The person may lose mass or suffer from deficiencies, and more unabsorbed food may be passed out in the faeces.

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